Jaskaran Singh

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Education

Vellore Institute of Technology, Vellore — Vellore, TN
Bachelor of Technology in Computer Science and Engineering — *Jul 2026 – Present*

Amity International School, Noida — Noida, UP
Physics, Chemistry, Mathematics — *Mar 2024 – May 2026*

Experience

ARGhouse — Jul 2025 – Dec 2025
Security Researcher — *Remote*

- Contributed security testing for web assets and campaign infrastructure related to marketing initiatives for The Undertone.
- Conducted reconnaissance and web application security assessments.

Cicada Productions LLC — Feb 2025 – Apr 2025
Security Researcher — *Remote*

- Addressed page enumeration, SSRF, advanced XSS, clickjacking risks, and DDoS/DoS mitigation vulnerabilities.

Projects

Paladin | *Python, TypeScript, SentenceTransformers* — 2026

- Built a security layer for AI coding agents that checks tool calls before execution, helping prevent unsafe commands and unauthorized access to sensitive files.
- Used SentenceTransformers, cosine similarity, and rule-based checks to analyze agent actions and assign risk scores.
- Added a permission system that flags risky actions and requests user approval before allowing execution.
- Implemented action logging and file monitoring to track agent activity and investigate suspicious behavior.

RAG Lens | *Typescript, RAG, Embeddings* — 2026

- Engineered a browser-based RAG observability tool using Google Gemini, enabling real-time visualization of document relationships and retrieval decisions.
- Built an end-to-end document ingestion and retrieval pipeline supporting PDF, DOCX, and TXT, with embedding-based search, reranking, and source-level citations.
- Developed interactive knowledge graph visualization and pipeline tracing, exposing semantic similarity scores, retrieved passages, and per-stage latency for debugging RAG responses.

Achievements

- Qualified for the final round of **Code2Create 7.0 (2026)**, held at VIT Vellore, and **NexHack 2.0 (2025)**, held at IITM.
- Won the **Best Music Award** at **Daydream 2025**, a hackathon held at IIT Delhi.
- Won **2nd place** in **Encryptid CTF 2025**, held online.

Technical Skills

Languages: Python, C++, Nim, R
Frameworks: FastAPI, LangChain, LangGraph
AI Security: Prompt Injection Defense, Agent Security, Tool-Call Validation, LLM Guardrails
Libraries: Pandas, NumPy, scikit-learn, PyTorch, Hugging Face Transformers, SentenceTransformers
Other Tools: Git, GitHub, Docker, Linux, PostgreSQL, Gemini API, Cloudflare, Vercel